慧脉守护

Huimai Guardian

Ward Intelligent Agent Platform & Service Robot System

Bilibili Video: https://b23.tv/tSfsfaY

Youtube Video: https://youtu.be/sfT0goJFcDs

GitHub: https://github.com/ZXZ12310304/wardlung-composs-v2

Team Members: Yu Jia, Zhang Xinzhu, Jin Aoqing, Liu Yuntian, Pang Yuhui

Background

Data Source: National Bureau of Statistics

慧脉守护

Rising Service Pressure → Growing Demand for Smart Wards

Meducal Pressure

In Q1 2026, total patient visits at medical and healthcare institutions nationwide reached 1.86 billion.
Total hospital discharges reached 75.522 million, with a bed utilization rate of 78.8%.







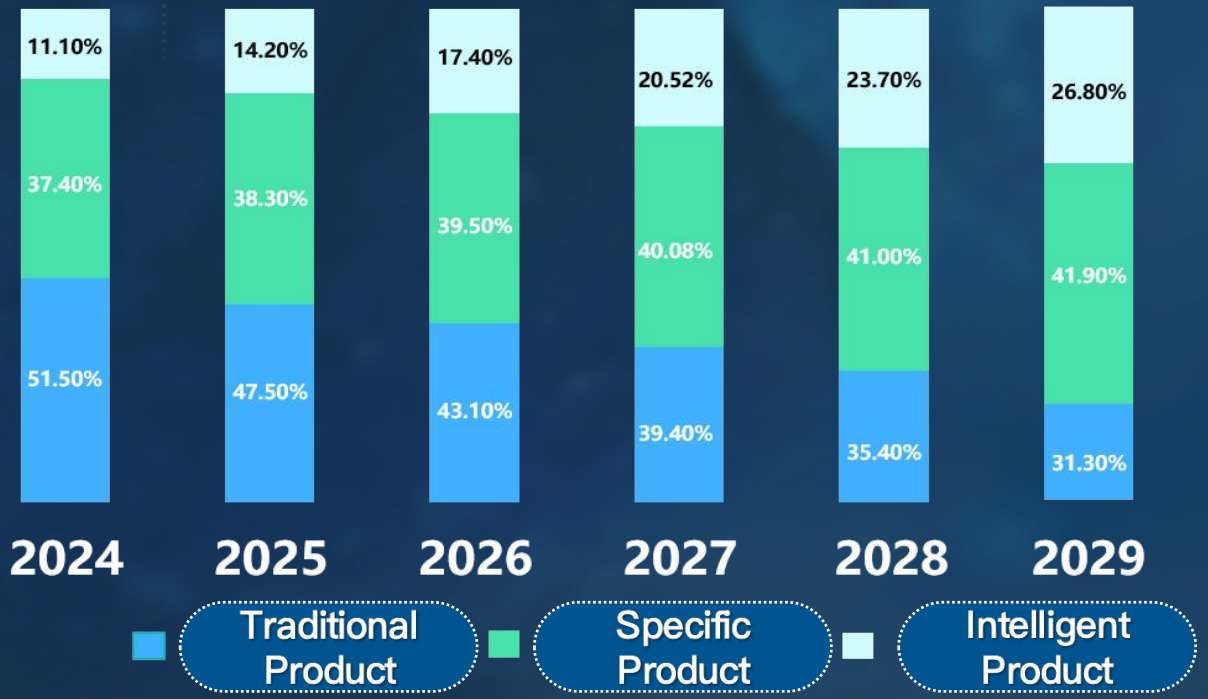
The CPC Central Committee and State Council prioritize healthcare.

Market Analysis

慧脉守护

Accelerated Construction of Smart Wards, AI + Medical Collaboration Emerges as a New Growth Direction

Product Structure and Proportion of China's Medical Device Market (2024 Actual –2029 Forecast)



In the future, the proportion of traditional medical devices will decrease, while the proportion of special scenario and intelligent products will increase.

Timeliness of Emergency ResponseThe inpatient population is growing continuously, placing higher demands on the handling of emergency events.

Intelligence of Medical PlatformsThere is a greater demand to reduce the probability of human error and achieve minimal-staff intelligent operations.

Diversification of Disposal MethodsWard demands are complex and variable, requiring synchronous improvement at the processing end.

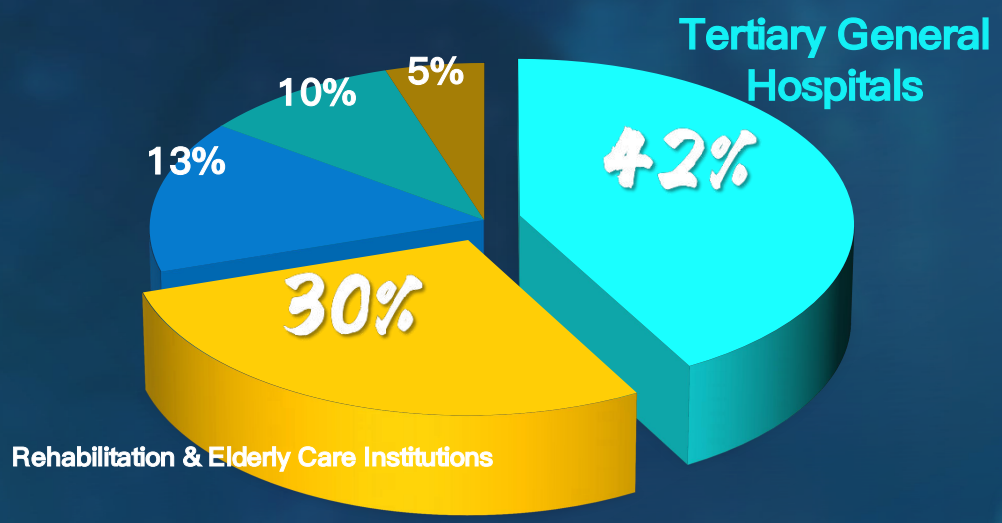
Market Analysis

慧脉守护

Broad customer base, dominated by tertiary general hospitals, accounting for 70%

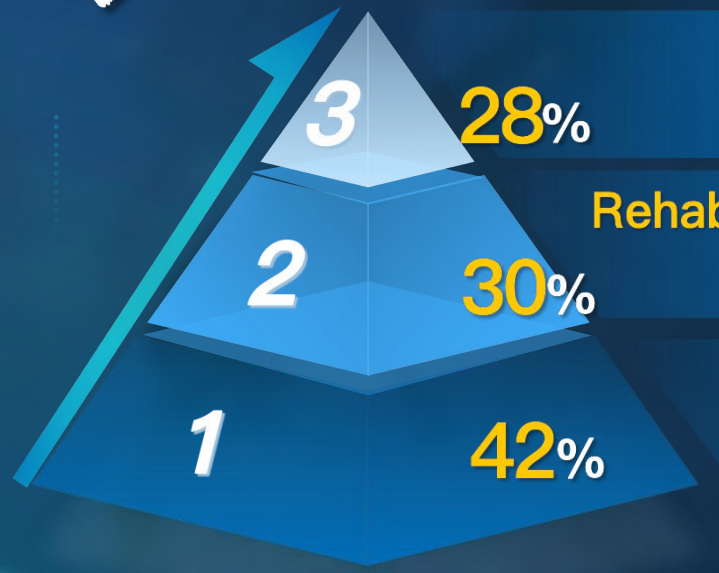
S

Segmented by customer type and industry sector



T

Target Market Selection



Other Medical Institutions

Rehabilitation and Elderly Care Institutions

Tertiary General Hospitals

P

Hospitals

The Second Affiliated Hospital of Soochow University

Shanghai Jiao Tong University Affiliated Medical Center

Hospital Needs

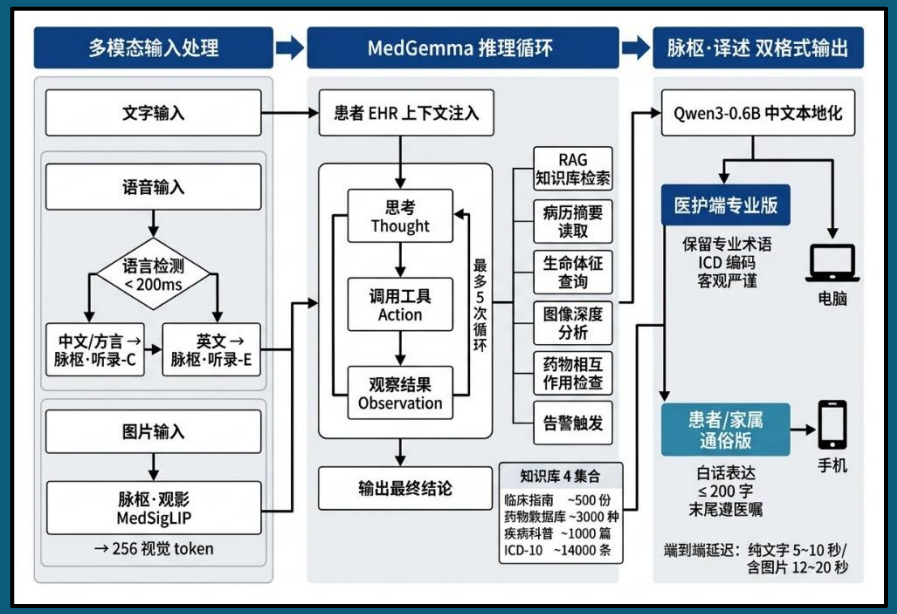
Alleviate the heavy burden on frontline medical staff in high-frequency patient response and manual ward rounds.

Product System

慧脉守护

Based on a multimodal medical AI agent engine, it enhances the efficiency of ward inquiry and auxiliary analysis, achieving unified understanding of text, voice, and images.

Multimodal Input Capability



The Huimai AI Agent Engine supports three types of input: text, voice, and medical images. It can integrate patient medical records, clinical knowledge bases, and current context to uniformly process inquiries, symptom expressions, and auxiliary analysis requests in the ward.

Knowledge-Enhanced Reasoning

ReAct + EHR +
Knowledge Base
Collaborative Analysis

Through "multimodal input + knowledge-enhanced reasoning + dual-format output," the Huimai AI Agent Engine significantly improves the efficiency of ward inquiry, auxiliary analysis, and doctor-patient communication.

Product System

慧脉守护

Four-Terminal Collaborative Connectivity: Building a Closed-Loop Smart Ward Service System

Patient Side



Supports symptom reporting, voice/text Q&A, and access to key care instructions, helping patients express their needs more conveniently, understand their current condition, and receive daily care reminders

Family Side



Supports remote access to patient updates, informed consent signing, alert notifications, and key information feedback, reducing information asymmetry for family members and enhancing their sense of involvement and reassurance.

Nurse Side



Supports centralized monitoring of ward patients' status, abnormal vital signs, and pending tasks, helping nurses quickly identify risks, improve ward-round efficiency, and reduce repetitive communication workload.

Doctor Side



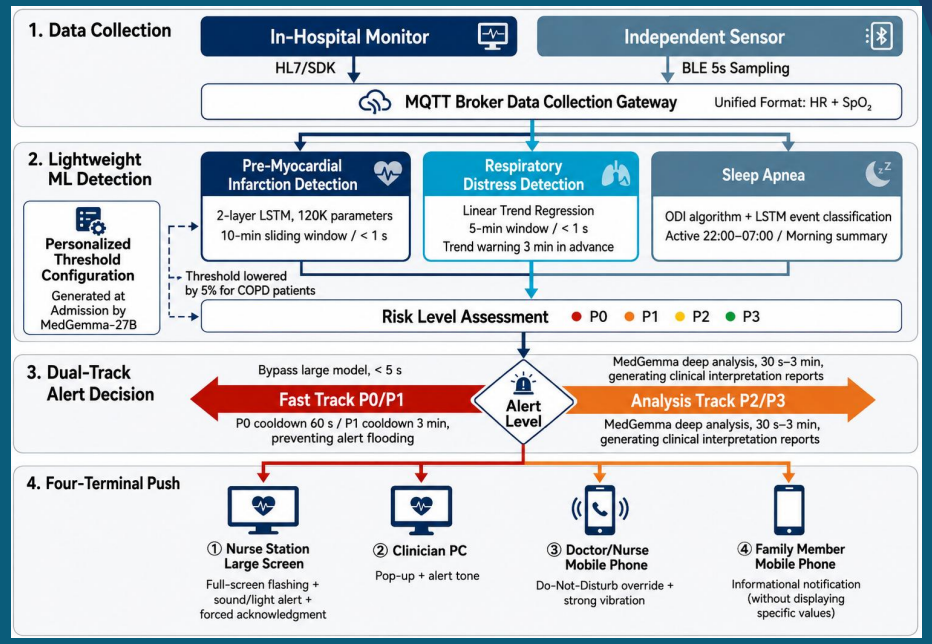
Supports centralized access to patient information, AI-assisted analysis results, and risk alerts, helping doctors complete evaluations, communication, and follow-up diagnosis and treatment decisions more efficiently.

Product System

慧脉守护

Continuously collect key vital-sign data such as HR and SpO_2 , and identify high-risk conditions in real time, including early signs of myocardial infarction, respiratory distress, and nighttime sleep apnea, thereby improving the ward's ability to detect risks in advance.

Real-Time Monitoring Scenario



Abnormal Vital-Sign Detection

The system continuously monitors key bedside vital signs such as heart rate and blood oxygen saturation in the ward. It covers high-risk scenarios including early signs of myocardial infarction, respiratory distress, and nighttime sleep apnea, enabling real-time detection and early identification of abnormal conditions.

Effect Comparison

Scenario	Traditional Method	Huimai Guardian
Abnormality Detection	Scheduled ward rounds, relying on manual checks	Real-time continuous monitoring
High-Risk Response	Detection may be delayed	P0 critical alerts within 5 seconds
Moderate Abnormality Handling	Scattered information, relying on experience-based judgment	AI-generated clinical interpretation reports



Two application validation certificates have been obtained, demonstrating that partner institutions recognize the project's practical application value.

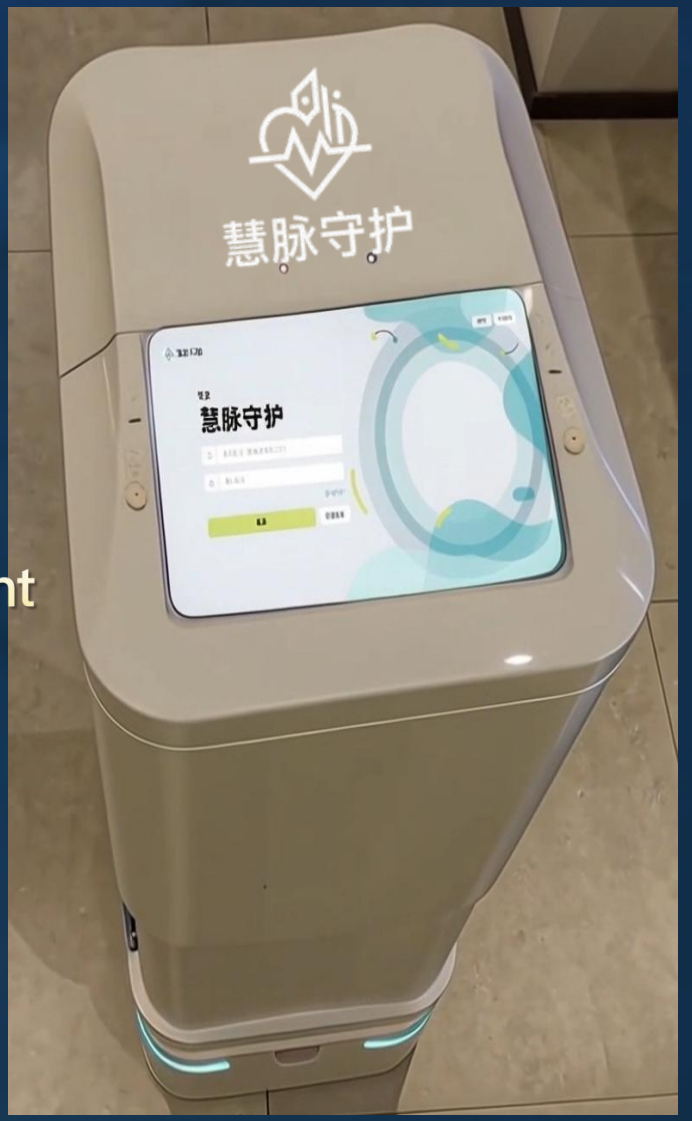
Based on continuous vital-sign monitoring and a dual-track alert mechanism, the system builds a ward risk early-warning platform that enables instant identification and rapid notification of high-risk abnormalities. P0-level critical events can be delivered to all four terminals within 5 seconds。.

慧脉守护

Main Hardware Configuration of the Huimai Robot

Category	Equipment Name
Mobile Chassis	AgileX Ranger Mini 3.0
Mechanical Structure	MISUMI 40×40 Aluminum Profile, Customized Stainless-Steel Tray
Main Navigation Perception	SICK TiM781S / Hokuyo UST-20LX
Short-Range Visual Perception	Intel RealSense D455
Auxiliary Vision	Logitech Brio 4K
Display Interaction	13.3-inch Capacitive Touchscreen
Voice Interaction	reSpeaker USB 4-Mic Array
Storage Module	Samsung 980 PRO 2TB NVMe SSD
Communication Module	Intel Wi-Fi 6E AX210
Medical Data Access	VitalPatch / Nonin WristOx2 / Philips IntelliVue MX450
Safety Control	Schneider Electric Emergency Stop Button, Auxiliary Power

Huimai Guardian Robotic Deployment

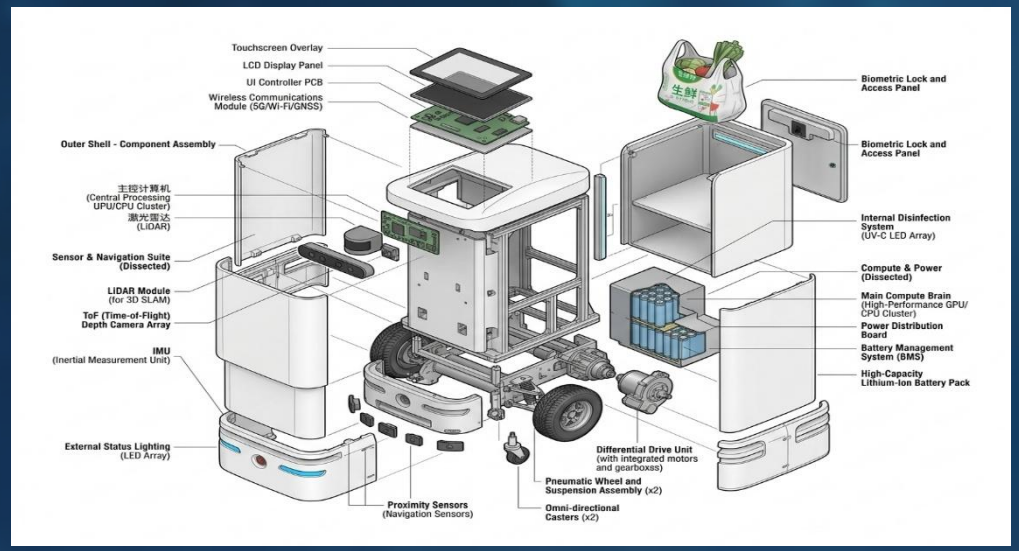


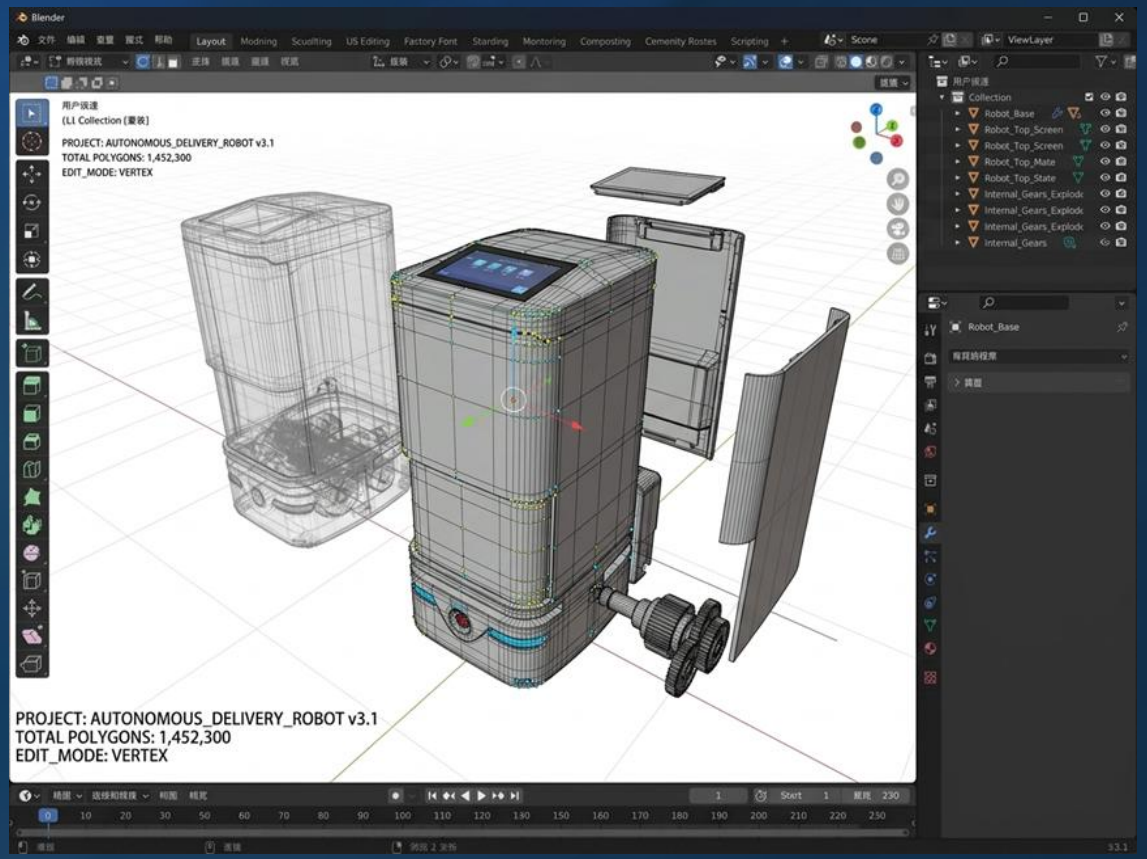
Business Plan

慧脉守护

Product R&D Upgrades and Improved Industrial Services

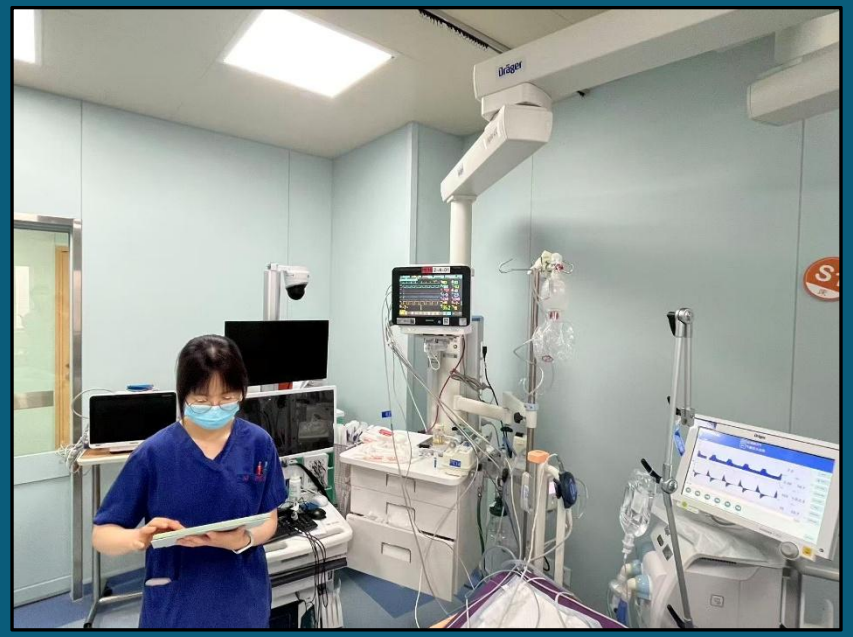
Expand into long-term care scenarios such as elderly care institutions and rehabilitation centers, and integrate wearable devices to extend continuous monitoring capabilities beyond hospitals.





In the maturity stage, the system will explore compliant integration with medical insurance data, support the continuous iterative optimization of inpatient risk prediction models, and gradually build an intelligent healthcare service system covering the full cycle of hospitalization, rehabilitation, and home care.

Throughon-site testing and real-world application demonstrations in hospital settings,, the platform's interaction processes and practical implementation capabilities are intuitively validated within actual ward environments.



The project has completed the application demonstration and functional verification in a real hospital environment, covering core aspects such as ward interaction, doctor-nurse collaboration, and system operation.

Hospital application verification

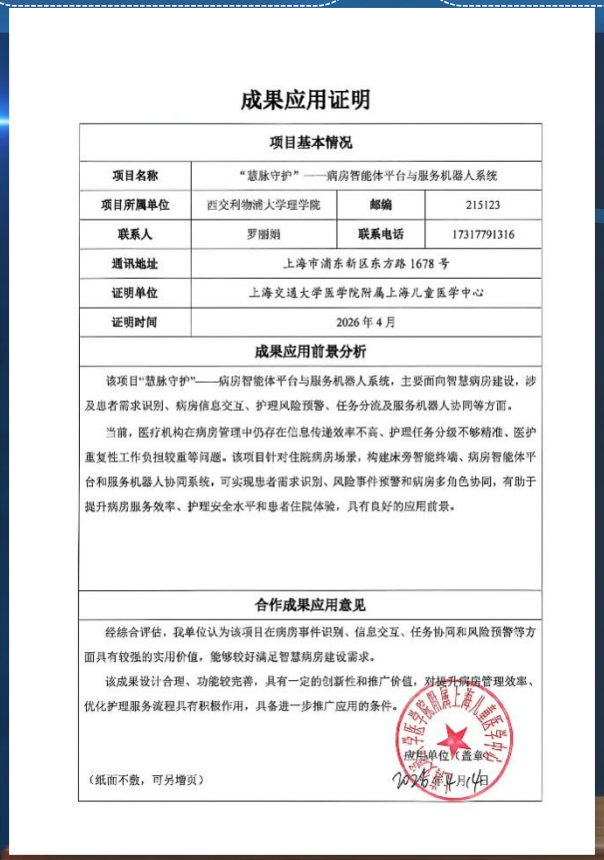




Through the real-scene testing and application demonstration of the hospital, the project has comprehensively verified the interactive feasibility, collaborative ability and practical value of the intelligent platform for hospital wards in real scenarios.

Close to the hospital ward close to the medical staff close to the application

Hospital on-site application



成果应用证明

项目基本情况			
项目名称	“慧脉守护”——病房智能体平台与服务机器人系统		
项目所属单位	西交利物浦大学理学院	邮编	215123
联系人	罗丽娟	联系电话	17317791316
通讯地址	上海市浦东新区东方路 1678 号		
证明单位	上海交通大学医学院附属上海儿童医学中心		
证明时间	2026 年 4 月		

成果应用前景分析

该项目“慧脉守护”——病房智能体平台与服务机器人系统，主要面向智慧病房建设，涉及患者需求识别、病房信息交互、护理风险预警、任务分流及服务机器人协同等方面。

当前，医疗机构在病房管理中仍存在信息传递效率不高、护理任务分级不够精准、医护重复性工作负担较重等问题。该项目针对住院病房场景，构建床旁智能终端、病房智能体平台和服务机器人协同系统，可实现患者需求识别、风险事件预警和病房多角色协同，有助于提升病房服务效率、护理安全水平和患者住院体验，具有良好的应用前景。

合作成果应用意见

经综合评估，我单位认为该项目在病房事件识别、信息交互、任务协同和风险预警等方面具有较强的实用价值，能够较好满足智慧病房建设需求。

该成果设计合理、功能较完善，具有一定的创新性和推广价值，对提升病房管理效率、优化护理服务流程具有积极作用，具备进一步推广应用的条件。

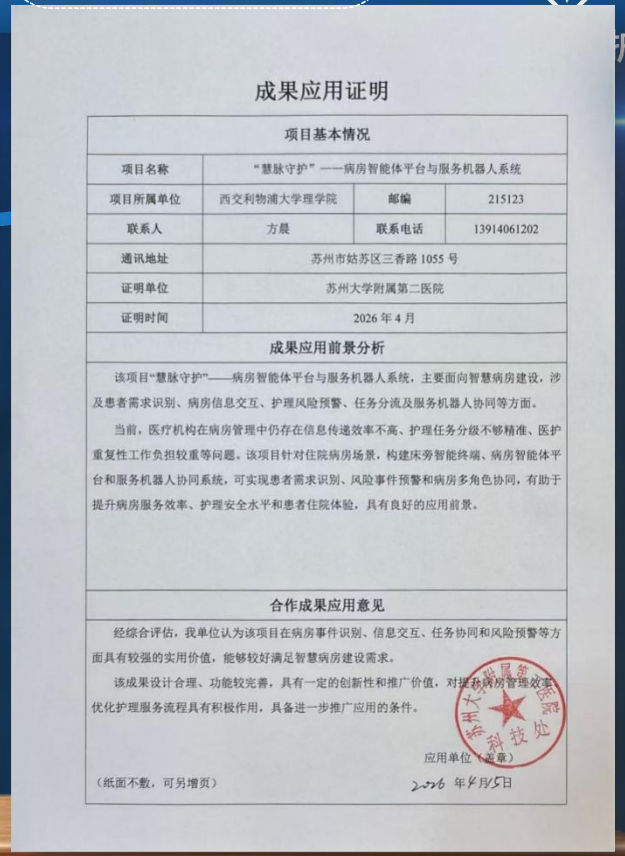
应用单位（盖章）

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2026 年 4 月 4 日

1.The Second Affiliated Hospital of Soochow University

2. Shanghai Children's Medical Center, Affiliated to Shanghai Jiao Tong University School of Medicine



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联系人	方晨	联系电话	13914061202
通讯地址	苏州市姑苏区三香路 1055 号		
证明单位	苏州大学附属第二医院		
证明时间	2026 年 4 月		

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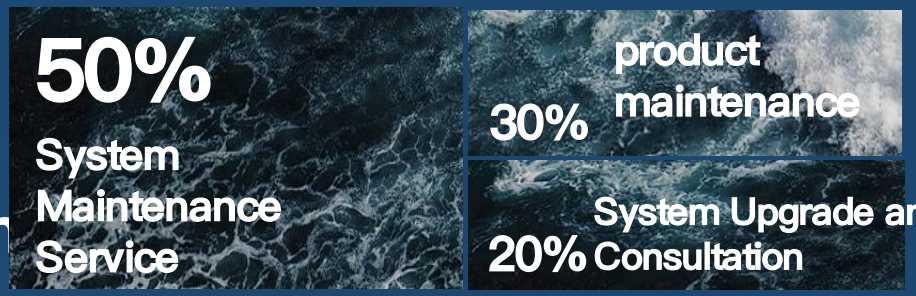
The application results have proved that our products are of significant help in hospital ward management.

The application of the results has been proven.

The business chain is well-established and holds great potential for profit.

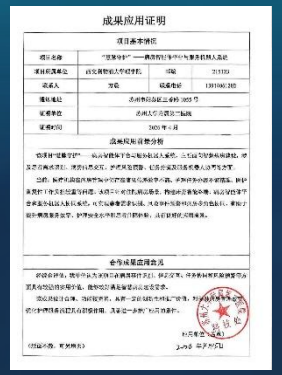
product

It is expected that each unit of the equipment will be sold for20,000.
The system and equipment are sold for50,000, and th
service fee is300,000.





The Second Affiliated Hospital of Soochow University


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联系人	罗丽娟	联系电话	17317791316
通讯地址	苏州市经济技术开发区仁爱路 111 号		
证明单位	苏州大学附属第二医院		
证明时间	2025 年 9 月		

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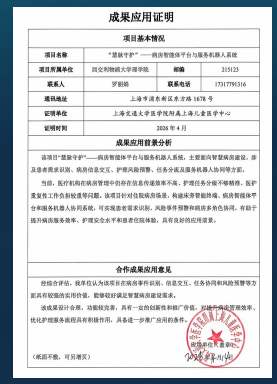
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慧脉守护

Direct Sales

Core tech licensing
Partial tech transfer
Individual /bundled sales



Agent Sales

Delivery to agents
OEM partner manufacturing
Distribution & IP licensing



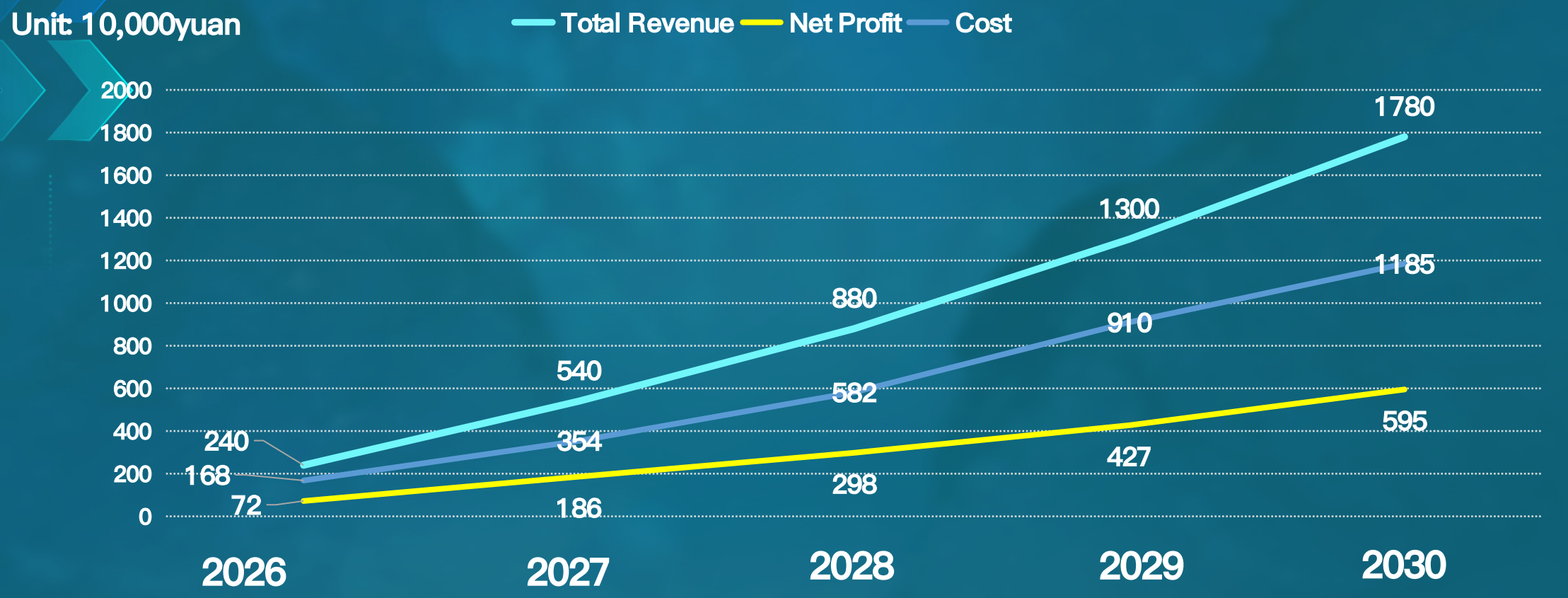
After-sales Service

Training & installation
Follow-up visits
Repair & reuse



慧脉守护

Diversified development of services, with considerable profitability and efficiency

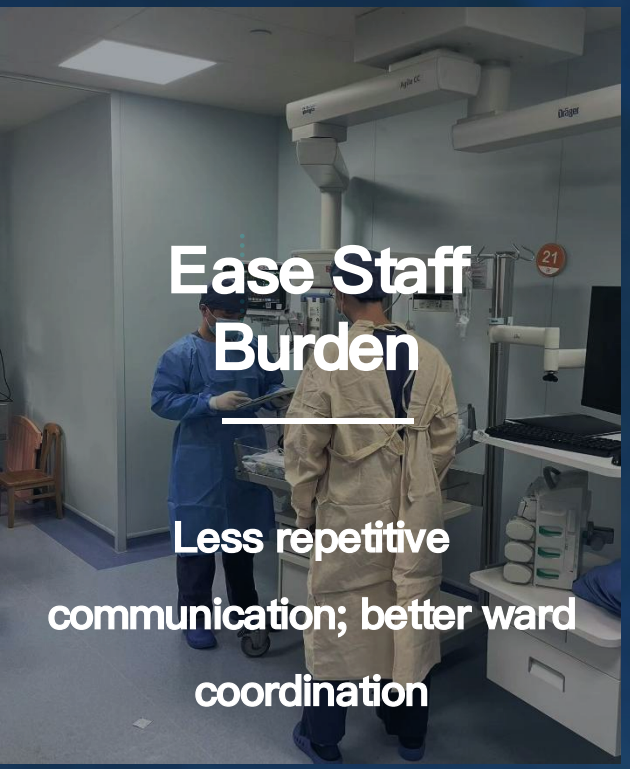


Payback Period
2.37 years

Total Profit
RMB 17.8M

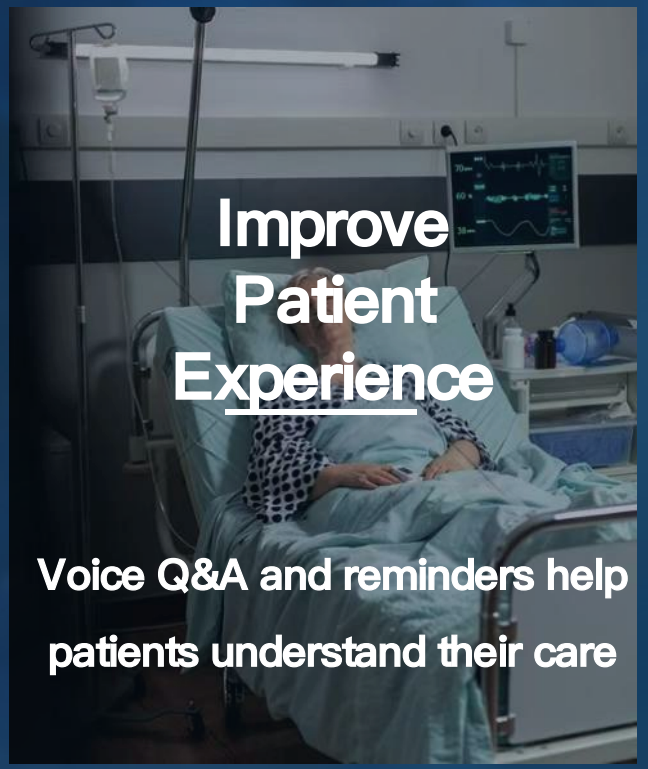
Net Profit
RMB 5.95M

Social Benefits: efficient wards, safer patients, smarter healthcare



Ease Staff Burden

Less repetitive communication; better ward coordination



Improve Patient Experience

Voice Q&A and reminders help patients understand their care



Protect Ward Safety

Vital-sign alerts help staff detect risks earlier



Upgrade Smart Healthcare

Digital ward services support smart hospital deployment

Patient-centered collaboration: more efficient, safer, smarter ward services

Win in Suzhou · Create the Future
Global Entrepreneurship Competition

慧脉守护

Huimai Guardian

Ward Agent Platform and Service Robot System